Technical Document #989: Data Engineering - Comprehensive Overview

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Data lakes store raw data in its native format. They support structured, semi-structured, and unstructured data at any scale.

ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a target data store.

Data quality ensures data is accurate, complete, and consistent. Data validation, cleansing, and monitoring are essential components.

Data governance establishes policies and procedures for managing data assets. It covers data privacy, security, and compliance.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Key Takeaways:

- Partitioning divides data into smaller, more manageable pieces.
- Data lineage tracks the origin and movement of data through systems.
- Data lakes store raw data in its native format.